ClipSAT : Cambridge AS Level Math 9709 [ANSWER KEY]

Unit 0.01: AS-Level 9709 Paper 1/2/4/5 Diagnostic Readiness

Q1. The diagram shows a right-angled triangle ABC, with the right angle at D. The altitude from D to the hypotenuse BC is drawn. Given that BD = 4 cm and DC = 9 cm, find the length of the altitude AD.

[x] A: 6 cm

[ ] B: 5 cm

[ ] C: 7 cm

[ ] D: 8 cm

Solution: **Mark Scheme CAIE / Edexcel Grading Steps:** - **Step 1 Method & Formulation:** Identify the required geometric/algebraic relationship from the diagram and substitute the given values. **[M1]** - **Step 2 Derivation & Calculation:** In a right-angled triangle, the altitude to the hypotenuse creates three similar triangles. Specifically, triangle ABD is similar to triangle CAD, and both are similar to the original triangle ABC. By the geometric mean theorem or by using similarity ratios, the square of the altitude to the hypotenuse is equal to the product of the two segments it divides the hypotenuse into.
Therefore, AD^2 = BD times DC.
Given BD = 4 cm and DC = 9 cm.
AD^2 = 4 times 9
AD^2 = 36
AD = sqrt{36}
AD = 6 cm. **[M1]** - **Step 3 Final Accuracy:** Verified exact result: 6 cm. **[A1]**

Q2. A fair six-sided die is rolled 4 times. Let X be the number of times a 6 is obtained. The possible values for X are 0, 1, 2, 3, and 4. The probability of obtaining a 6 on any given roll is p = frac{1}{6} . The number of trials is n=4 . This scenario can be modelled by a binomial distribution, denoted as X sim Bn, p . Which of the following represents the correct parameters for this binomial distribution?

[x] A: B4, 1/6

[ ] B: B6, 1/4

[ ] C: B1, 4/6

[ ] D: B4, 4/6

Solution: The problem describes a situation with a fixed number of independent trials rolling a die 4 times, where each trial has only two possible outcomes obtaining a 6 or not obtaining a 6, and the probability of success obtaining a 6 is constant for each trial. In a binomial distribution, denoted as X sim Bn, p , n represents the number of trials, and p represents the probability of success on a single trial. From the question: - The number of times the die is rolled is 4, so n = 4 . - The probability of obtaining a 6 on any given roll is p = frac{1}{6} . Therefore, the binomial distribution is correctly represented as X sim B4, frac{1}{6} . [M1][A1]

Q3. The diagram shows part of the curve with equation y = x^2 - 4x + 5 and the point P3, 2 lying on the curve. Find the gradient of the tangent to the curve at the point P .

[x] A: 2

[ ] B: 6

[ ] C: -4

[ ] D: dfrac{2}{3}

Solution: **Mark Scheme CAIE / Edexcel Grading Steps:** - **Step 1 Method & Formulation:** Identify the required geometric/algebraic relationship from the diagram and substitute the given values. **[M1]** - **Step 2 Derivation & Calculation:** To find the gradient of the tangent to the curve at P3, 2 , we determine the derivative dfrac{mathrm{d}y}{mathrm{d}x} and evaluate it at x = 3 . 1. Differentiate y = x^2 - 4x + 5 with respect to x using the power rule: frac{mathrm{d}y}{mathrm{d}x} = 2x - 4 2. Substitute x = 3 into the derivative: left.frac{mathrm{d}y}{mathrm{d}x}right|_{x=3} = 23 - 4 = 6 - 4 = 2 Common errors: - 6 occurs if only the first term is differentiated giving 2x and evaluated at x = 3 , omitting the derivative of -4x . - -4 incorrectly takes the linear coefficient without differentiating the quadratic term. - dfrac{2}{3} confuses the instantaneous gradient of the tangent with the gradient of the line from the origin to P , dfrac{y}{x} = dfrac{2}{3} . **[M1]** - **Step 3 Final Accuracy:** Verified exact result: 2 . **[A1]**

Q4. The diagram shows a triangle ABC with coordinates A2, 1, B8, 1 and C5, 5. Find the length of the side AC.

[x] A: 5
[ ] B: sqrt{26}
[ ] C: 6
[ ] D: 7

Solution: **Mark Scheme CAIE / Edexcel Grading Steps:** - **Step 1 Method & Formulation:** Identify the required geometric/algebraic relationship from the diagram and substitute the given values. **[M1]** - **Step 2 Derivation & Calculation:** The coordinates of A are 2, 1 and the coordinates of C are 5, 5. Using the distance formula, d = sqrt{x_2 - x_1^2 + y_2 - y_1^2}. Length of AC = sqrt{5 - 2^2 + 5 - 1^2} Length of AC = sqrt{3^2 + 4^2} Length of AC = sqrt{9 + 16} Length of AC = sqrt{25} Length of AC = 5. **[M1]** - **Step 3 Final Accuracy:** Verified exact result: 5. **[A1]**

Q5. The diagram shows a right-angled triangle ABC, with the right angle at D. The altitude from D to the hypotenuse BC is drawn. Given that BD = 4 cm and DC = 9 cm, find the length of the altitude AD.

[x] A: 6 cm
[ ] B: 5 cm
[ ] C: 7 cm
[ ] D: 8 cm

Solution: **Mark Scheme CAIE / Edexcel Grading Steps:** - **Step 1 Method & Formulation:** Identify the required geometric/algebraic relationship from the diagram and substitute the given values. **[M1]** - **Step 2 Derivation & Calculation:** In a right-angled triangle, the altitude to the hypotenuse creates three similar triangles. Specifically, triangle ABD is similar to triangle CAD, and both are similar to the original triangle ABC. By the geometric mean theorem or by using similarity ratios, the square of the altitude to the hypotenuse is equal to the product of the two segments it divides the hypotenuse into.
Therefore, AD^2 = BD times DC.
Given BD = 4 cm and DC = 9 cm.
AD^2 = 4 times 9
AD^2 = 36
AD = sqrt{36}
AD = 6 cm. **[M1]** - **Step 3 Final Accuracy:** Verified exact result: 6 cm. **[A1]**

Q6. A fair six-sided die is rolled 4 times. Let X be the number of times a 6 is obtained. The possible values for X are 0, 1, 2, 3, and 4. The probability of obtaining a 6 on any given roll is p = frac{1}{6} . The number of trials is n=4 . This scenario can be modelled by a binomial distribution, denoted as X sim Bn, p . Which of the following represents the correct parameters for this binomial distribution?

[x] A: B4, 1/6
[ ] B: B6, 1/4
[ ] C: B1, 4/6
[ ] D: B4, 4/6

Solution: The problem describes a situation with a fixed number of independent trials rolling a die 4 times, where each trial has only two possible outcomes obtaining a 6 or not obtaining a 6, and the probability of success obtaining a 6 is constant for each trial. In a binomial distribution, denoted as X sim Bn, p , n represents the number of trials, and p represents the probability of success on a single trial. From the question: - The number of times the die is rolled is 4, so n = 4 . - The probability of obtaining a 6 on any given roll is p = frac{1}{6} . Therefore, the binomial distribution is correctly represented as X sim B4, frac{1}{6} . [M1][A1]

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Q7. The diagram shows part of the curve with equation y = x^2 - 4x + 5 and the point P3, 2 lying on the curve. Find the gradient of the tangent to the curve at the point P .

[x] A: 2

[ ] B: 6

[ ] C: -4

[ ] D: dfrac{2}{3}

Solution: **Mark Scheme CAIE / Edexcel Grading Steps:** - **Step 1 Method & Formulation:** Identify the required geometric/algebraic relationship from the diagram and substitute the given values. **[M1]** - **Step 2 Derivation & Calculation:** To find the gradient of the tangent to the curve at P3, 2 , we determine the derivative dfrac{mathrm{d}y}{mathrm{d}x} and evaluate it at x = 3 . 1. Differentiate y = x^2 - 4x + 5 with respect to x using the power rule: frac{mathrm{d}y}{mathrm{d}x} = 2x - 4 2. Substitute x = 3 into the derivative: left.frac{mathrm{d}y}{mathrm{d}x}right|_{x=3} = 23 - 4 = 6 - 4 = 2 Common errors: - 6 occurs if only the first term is differentiated giving 2x and evaluated at x = 3 , omitting the derivative of -4x . - -4 incorrectly takes the linear coefficient without differentiating the quadratic term. - dfrac{2}{3} confuses the instantaneous gradient of the tangent with the gradient of the line from the origin to P , dfrac{y}{x} = dfrac{2}{3} . **[M1]** - **Step 3 Final Accuracy:** Verified exact result: 2 . **[A1]**

Q8. The diagram shows a triangle ABC with coordinates A2, 1, B8, 1 and C5, 5. Find the length of the side AC.

[x] A: 5

[ ] B: sqrt{26}

[ ] C: 6

[ ] D: 7

Solution: **Mark Scheme CAIE / Edexcel Grading Steps:** - **Step 1 Method & Formulation:** Identify the required geometric/algebraic relationship from the diagram and substitute the given values. **[M1]** - **Step 2 Derivation & Calculation:** The coordinates of A are 2, 1 and the coordinates of C are 5, 5. Using the distance formula, d = sqrt{x_2 - x_1^2 + y_2 - y_1^2}. Length of AC = sqrt{5 - 2^2 + 5 - 1^2} Length of AC = sqrt{3^2 + 4^2} Length of AC = sqrt{9 + 16} Length of AC = sqrt{25} Length of AC = 5. **[M1]** - **Step 3 Final Accuracy:** Verified exact result: 5. **[A1]**